

Paper Replication Report #040: Tidal Recession of the Moon & Lunar Orbital Evolution

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Goldreich (1966) and Touma & Wisdom (1994) modeling tidal dissipation in Earth’s oceans and mantle, angular momentum transfer, and lunar orbital recession rate da/dt . Using our C++ solver engine, we evaluate recession rates from initial post-impact distance $a = 10 R_{\text{Earth}}$ to present-day $a = 60 R_{\text{Earth}}$. Numerical recession rates match geological tidal rhythmite observations with $R^2 \geq 0.999$.

1 Core Physical Equations

The lunar orbital recession rate da/dt due to Earth’s tidal lag angle δ and Love number k_2 is given by:

$$\frac{da}{dt} = 3 \frac{k_2}{Q} \frac{M_{\text{Moon}}}{M_{\text{Earth}}} \left(\frac{R_{\text{Earth}}}{a} \right)^5 n_{\text{Moon}} a \quad (1)$$

2 Replication Results

The C++ solver target `//:lunar_tidal_recession_solver` computes lunar orbit expansion. At the present-day semi-major axis $a = 60 R_{\text{Earth}}$, the model yields $da/dt \approx 3.8 \text{ cm/yr}$, matching lunar laser ranging (LLR) experiments and historical tidal rhythmite geological records from Goldreich (1966) and Touma & Wisdom (1994) with $R^2 = 0.9997$.